IP

Indresh Pathak

Senior ML/MLOps | GenAI & Agentic AI Architect | 12+ Years

CONTACT

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CORE TECHNICAL SKILLS

LANGUAGES & SQL

- Python (8+ yrs) • SQL (BigQuery)
- Node.js • TypeScript • Java

ML / MLOPS / GENAI

- Vertex AI • MLflow • KubeFlow
- TensorFlow • PyTorch • Scikit-learn
- Generative AI • LLMs • RAG Pipelines
- LangChain • LangGraph • Agentic AI
- Prompt Engineering • Fine-Tuning

Indresh Pathak

Senior ML/MLOps Engineer | GenAI & Agentic AI Architect | Solution Architect

Professional Summary

Results-driven **Senior Software Engineer and ML/MLOps Architect with 12+ years of experience** building and deploying production-grade machine learning systems, GenAI solutions, and scalable cloud-native platforms across e-commerce, EdTech, healthcare, and retail domains. Deep hands-on expertise in **Python, Node.js, ML frameworks (TensorFlow, PyTorch, Scikit-learn), MLOps tooling (Vertex AI, MLflow, KubeFlow)**, and **Generative AI (LLMs, RAG pipelines, LangChain, LangGraph, Agentic AI, prompt engineering)**. Proven track record of building autonomous multi-agent AI systems, retrieval-augmented generation (RAG) workflows, and end-to-end ML lifecycle management—from feature engineering and model training through automated CI/CD deployment, model registry, and production monitoring. Extensive experience with **GCP, CockroachDB, distributed databases**, **DevOps pipelines (Docker, Kubernetes, Terraform, GitHub Actions)**, and Agile methodologies. Adept at leading cross-functional teams of 10–25 engineers, driving digital transformation, and delivering measurable business outcomes including **40% latency reduction, 3× throughput improvements, and 35% faster deployment cycles**.

Professional Experience

Senior Technology Solution Architect

Broadwayinfotech, Noida, India — *2024 – Present*

Architecting ML-powered event booking platform and GenAI-integrated omnichannel CRM with end-to-end MLOps pipelines on GCP

- **GenAI-Powered Recommendations (LangChain + RAG):** Designed and deployed a **Generative AI pipeline** using LangChain, LLMs, and RAG architecture for dynamic product descriptions and AI/ML-based event recommendations, increasing conversion rates by 28% and upsell revenue by 18%
- **MLOps Pipeline (Vertex AI + KubeFlow):** Built end-to-end ML pipelines on **GCP Vertex AI** with KubeFlow orchestration for model training, versioning (Model Registry), automated retraining, and production deployment — reducing model-to-production time by 60%
- **Real-Time ML Inference:** Deployed real-time ML inference services using **FastAPI + Docker + Kubernetes**, serving 50K+ predictions/day with p99 latency under 120ms

Model Registry • Feature Store

Model Monitoring (Drift & Perf.)

CLOUD & DATA

GCP (BigQuery, GCS, Dataflow, Dataproc)

AWS • Azure • Snowflake • Databricks

CockroachDB • PostgreSQL • MongoDB

BACK-END FRAMEWORKS

FastAPI • Flask • Node.js / Express.js

NestJS • Spring Boot • gRPC

DEVOPS & DEPLOYMENT

Docker • Kubernetes • Terraform (IaC)

CI/CD (GitHub Actions, Jenkins, GitLab)

SonarQube • FOSSA • 42Crunch

Tekton • Grafana • Prometheus

ARCHITECTURE & MESSAGING

Microservices • Distributed Systems

Kafka • Redis • Event-Driven Arch.

METHODOLOGIES

Agile / Scrum / Kanban

SDLC • TDD • Code Reviews

ML Lifecycle Management

EDUCATION

- **Omnichannel CRM with AI:** Implemented an intelligent omnichannel communication layer (email, WhatsApp, SMS, chat) with GenAI-powered auto-responses and sentiment analysis, handling 2M+ messages/month
- **DevOps & CI/CD:** Established **GitHub Actions CI/CD pipelines** with Docker containerization, Kubernetes orchestration, Terraform IaC, and automated quality gates (SonarQube, FOSSA) — achieving 35% faster deployment cycles
- **Model Monitoring:** Implemented production model monitoring dashboards (Grafana + Prometheus) for **drift detection and performance tracking**, ensuring model health with automated alerting and retraining triggers

Python | Vertex AI | KubeFlow | LangChain | RAG | GenAI / LLMs | MLflow | FastAPI | Docker | Kubernetes | Terraform | GitHub Actions | BigQuery | GCP | Redis | Socket.IO

Senior Technology Solution Architect

Petco, USA (Remote) — *Nov 2023 – Jul 2024*

> **Led ML model deployment, MLOps pipeline automation, and GenAI integration for Petco's e-commerce personalization and demand forecasting systems**

- **ML Model Deployment (TensorFlow + Scikit-learn):** Developed and deployed production ML models for **product recommendations, predictive demand forecasting, and dynamic pricing** — driving 22% increase in average order value and reducing stockouts by 30%
- **MLOps Practices:** Established comprehensive **MLOps lifecycle** — continuous model training, A/B testing, automated deployment via **MLflow + Vertex AI**, Feature Store management, and model monitoring with drift/performance detection
- **GenAI Personalization (LangChain + Agentic AI):** Integrated **LangChain-powered Agentic AI** for personalized product descriptions, RAG-based intelligent search, and an autonomous multi-agent customer support system, improving customer engagement by 35%
- **Data Pipeline (GCP BigQuery + Dataflow):** Architected data pipelines on **GCP BigQuery and Dataflow** for ETL processing of 50M+ events/day, feeding ML feature stores and analytics dashboards
- **DevOps & Infrastructure:** Managed **Kubernetes clusters** with Terraform IaC, CI/CD via GitLab CI and GitHub Actions, and implemented security scanning (42Crunch, SonarQube) across 40+ microservices
- **Agile Leadership:** Led a team of 15 engineers using Agile/Scrum methodologies with 2-week sprint cycles, achieving 95% sprint velocity consistency

Python | TensorFlow | Scikit-learn | MLflow | Vertex AI | LangChain | Agentic AI | GenAI | BigQuery | Dataflow | Kubernetes | Terraform | Kafka | gRPC | GitLab CI

Engineering Manager / Technical Architect

Netreo.inc, USA (Remote) — *Jun 2022 – Nov 2023*

Master of Science
Information Technology
IIIT Gwalior — 2014

CERTIFICATIONS

PMP® Training
Master Project Academy

Scrum Master Training
Master Project Academy

Automation Solution Architect
UiPath Academy

Built predictive ML models and MLOps infrastructure for cloud resource monitoring, with automated model retraining and deployment pipelines

- **Predictive ML Models (PyTorch + Scikit-learn):** Developed time-series forecasting models to predict cloud resource load spikes with 92% accuracy, enabling proactive auto-scaling and reducing downtime by 45%
- **MLOps Pipeline:** Architected end-to-end **MLOps pipeline** with automated model training, validation, registry (MLflow), and deployment via **Docker + Kubernetes** — supporting continuous model improvement cycle
- **Feature Engineering (BigQuery + Dataproc):** Built feature engineering pipelines on **GCP BigQuery and Dataproc** for processing cloud telemetry data at scale, feeding real-time and batch ML models
- **CI/CD & DevOps:** Implemented **Tekton-based CI/CD pipelines** with Terraform IaC for infrastructure provisioning, automated testing, and zero-downtime deployments across multi-cloud environments
- **Team Leadership:** Managed a team of 12 engineers, conducting code reviews, mentoring on ML best practices, and driving Agile delivery with 30% improvement in sprint velocity

Python | PyTorch | Scikit-learn | MLflow | BigQuery | Dataproc | Docker | Kubernetes | Tekton | Terraform | Grafana | Prometheus

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Technical Architect

Classplus, Noida, India — *Jul 2021 – Jun 2022*

Deployed ML models for real-time student feedback and built GenAI-powered content generation for EdTech platform serving 100K+ users

- **ML for Real-Time Feedback (TensorFlow):** Built and deployed predictive models for instant assessment feedback during live sessions, improving student performance tracking accuracy by 40% and reducing instructor response time by 50%
- **GenAI Content Generation:** Implemented **Generative AI pipelines** for automated quiz generation, lesson summarization, and personalized learning path recommendations using LLMs
- **Cloud Migration & MLOps (GCP):** Led migration of ML workloads to **GCP (Vertex AI, GCS, Dataflow)**, implementing cost-effective training infrastructure with automated model versioning and deployment — reducing cloud costs by 35%
- **DevOps:** Established Docker-based containerization with Kubernetes orchestration and CI/CD pipelines for both application and ML model deployment

Python | TensorFlow | GenAI / LLMs | Vertex AI | GCS | Dataflow | Docker | Kubernetes | React | AWS

Engineering Lead

Nfxdigital, New Delhi, India — *Jan 2016 – Jul 2021*

Built ML-powered analytics platform and predictive models for survey optimization, serving 500K+ users

- **ML-Powered Analytics (Scikit-learn + Python):** Developed predictive and clustering models to analyze survey responses at scale, generating actionable insights that improved campaign targeting by 25% and user engagement by 40%
- **Data Pipelines & ETL:** Built **Python-based ETL pipelines** with SQL (BigQuery) for processing millions of survey records, enabling real-time analytics dashboards and ML feature extraction
- **DevOps & Deployment:** Implemented **Docker containerization** and CI/CD pipelines (Jenkins, GitHub Actions) for automated testing and deployment, reducing release cycles from weeks to 2 days
- **Agile Leadership:** Led teams of 8–15 engineers through Agile/Scrum, driving iterative delivery and cross-functional collaboration across 5+ years

Python | Scikit-learn | BigQuery | Docker | Jenkins | GitHub Actions | Node.js | MongoDB | Redis

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Senior Software Engineer

Collective Artist, New Delhi, India

Apr 2014 – Jan 2016

- **Python Application Development:** Developed and maintained complex web applications using **Python (Flask)** and RESTful APIs, reducing critical bug resolution time by 60%
- **Data-Driven Solutions:** Built analytics modules and automated reporting using **Python and SQL**, enabling data-driven decision-making for product teams
- **Agile Collaboration:** Contributed to sprint planning, daily stand-ups, and retrospectives in an Agile environment, consistently delivering features ahead of schedule

Python | Flask | SQL | REST APIs | Agile

Key Achievements & Impact

- **ML Model Deployment:** Deployed 15+ production ML models across recommendation, forecasting, and NLP use cases with 99.5% uptime SLA
- **GenAI Implementations:** Delivered 4 GenAI solutions (RAG pipelines, LangChain-powered agentic AI workflows, LLM chatbots, auto-content generation, intelligent search) driving measurable business KPIs
- **MLOps Maturity:** Elevated MLOps maturity from ad-hoc to Level 2 automation — implementing model registry, feature stores, automated retraining, and drift monitoring across organizations
- **DevOps Excellence:** Reduced deployment frequency from monthly to daily releases through Docker/Kubernetes/Terraform-based CI/CD pipelines with 100% infrastructure-as-code coverage

- **Cost Optimization:** Achieved $200K+ annual cloud cost savings through GCP resource optimization, spot instance strategies, and workload right-sizing
- **Team Leadership:** Managed and mentored cross-functional teams of 10–25 engineers across ML, backend, and DevOps disciplines, achieving consistent sprint velocity improvements

Featured Projects

Agentic RAG Document Intelligence Platform

Personal / Open Source

2024

> **Multi-agent AI system for intelligent document Q&A with autonomous tool-calling, retrieval-augmented generation, and persistent memory — built with LangGraph and CockroachDB**

- **Agentic AI Architecture (LangGraph):** Designed a **multi-agent orchestration system** using LangGraph with autonomous agents for document ingestion, semantic retrieval, summarization, and Q&A — enabling complex multi-step reasoning with 87% answer accuracy
- **RAG Pipeline (LangChain + OpenAI):** Built a production RAG pipeline using **LangChain**, vector embeddings (FAISS/Chroma), and LLM re-ranking — reducing hallucination rate by 65% compared to naive LLM querying
- **Distributed Storage (CockroachDB):** Used **CockroachDB** as the distributed SQL backend for storing document metadata, user sessions, and agent conversation history — achieving 99.99% availability with geo-replicated reads
- **REST API (Python FastAPI):** Exposed all agent capabilities via a **FastAPI** REST service with async endpoints, JWT auth, and rate limiting; containerised with Docker and deployed on Kubernetes

Python | LangChain | LangGraph | RAG | Agentic AI | LLMs (OpenAI / Gemini) | CockroachDB | FastAPI | FAISS | Docker | Kubernetes

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LLM-Powered Real-Time Customer Support Agent

Personal / Proof of Concept

2024

> **Autonomous customer support system using LLMs and LangChain tool-calling agents; Node.js WebSocket backend; CockroachDB for ticket and session persistence**

- **LangChain Agent (Python):** Implemented a **LangChain ReAct agent** with custom tools for order lookup, knowledge-base retrieval, escalation logic, and proactive follow-up — resolving 74% of tickets autonomously without human intervention
- **Node.js WebSocket Server:** Built a high-throughput **Node.js / Express** WebSocket gateway handling 10K+ concurrent sessions, routing messages between the frontend chat UI and the Python agent microservice via an event-driven queue

- **CockroachDB Backend:** Persisted all ticket state, agent reasoning traces, and user history in **CockroachDB**, leveraging its distributed transactions to ensure no data loss across multi-region deployments
- **GenAI Sentiment Routing:** Integrated an LLM-based sentiment classifier (LangChain) to automatically escalate frustrated customers to human agents, reducing escalation time by 55%

Python | LangChain | GenAI / LLMs | Node.js | Express.js | CockroachDB | WebSocket | Docker | Redis

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Scalable E-Commerce Catalog API with GenAI Enrichment

Personal / Open Source

2023

> **High-performance product catalog service with LLM-powered auto-description generation, semantic search, and globally distributed CockroachDB storage**

- **LLM Auto-Enrichment (Python + LangChain):** Used **LangChain chains** to automatically generate SEO-optimised product descriptions and attribute tags from raw product specs, enriching 50K+ SKUs at 10× the speed of manual copywriting
- **Node.js Catalog API:** Developed a **Node.js (NestJS)** RESTful and GraphQL API for product CRUD, filtered search, and batch import — sustaining 8K req/sec under load with Redis caching layer
- **CockroachDB Multi-Region Schema:** Designed a geo-partitioned **CockroachDB** schema with table locality optimised for regional read latency — achieving sub-30ms p99 reads across US and EU regions
- **Semantic Search (RAG):** Integrated a **RAG-based semantic search** layer using LangChain + pgvector-compatible embeddings for natural language product discovery, improving search relevance by 42%

Node.js | NestJS | Python | LangChain | RAG | LLMs | CockroachDB | GraphQL | Redis | Docker